

5G FUNDAMENTALS AND ARCHITECTURES (EC23712)

ASSIGNMENT-1

TEAM – 5

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Team 5: Implement variable load ramp and plot rate following the ramp

1. Objective

To implement a variable load ramp using iPerf3 by gradually increasing the offered traffic load from 10 Mbps to 60 Mbps in fixed intervals and to visualize the corresponding achieved throughput over time. The objective is to analyze whether the network throughput accurately follows the increasing offered load and to evaluate the stability of data transmission during the experiment.

2. Software and Tools Used

Software	Purpose
Ubuntu Linux (22.04)	Operating system used for conducting the experiment.
Open5GS	Provides the 5G Core Network environment.
MongoDB	Stores subscriber information for Open5GS.
iPerf3	Generates network traffic and measures throughput.
Python 3	Processes throughput data and generates graphs.
Matplotlib	Creates throughput visualization graphs.
Terminal	Executes Linux commands and runs iPerf3 experiments.

3. System Setup

- Operating System: Ubuntu Linux 22.04
- 5G Core: Open5GS
- Database: MongoDB
- Traffic Generator: iPerf3
- Programming Language: Python 3
- Graph Plotting Tool: Matplotlib
- Transport Protocol: UDP
- Server Address: 127.0.0.1 (Loopback Interface)
- Port Number: 5201
- Test Duration: 60 seconds
- Traffic Pattern: Variable Load Ramp
- Bandwidth Levels: 10 Mbps → 20 Mbps → 30 Mbps → 40 Mbps → 50 Mbps → 60 Mbps
- Sampling Interval: 1 second

The iPerf3 server was started in server mode, and the client generated UDP traffic with increasing bandwidth every 10 seconds. The throughput values were collected throughout the experiment and plotted using Python.

4. Procedure

1. Install Open5GS, MongoDB, and iPerf3 on Ubuntu Linux.
2. Start the Open5GS core network and verify that all services are running.
3. Launch the iPerf3 server using the default port (5201).
4. Open another terminal and execute the iPerf3 client.
5. Generate UDP traffic by gradually increasing the offered bandwidth from 10 Mbps to 60 Mbps.
6. Record throughput values at one-second intervals.
7. Save the experimental output.
8. Use Python and Matplotlib to plot the Variable Load Ramp graph.
9. Compare the offered bandwidth with the achieved throughput.
10. Analyze the network performance during the entire experiment.

5. Commands Used

i. Start iPerf3 Server

```
iperf3 -s
```

ii. Generate Variable Load Ramp Traffic

```
iperf3 -c 127.0.0.1 -u -b 10M -t 10 -i 1
```

```
iperf3 -c 127.0.0.1 -u -b 20M -t 10 -i 1
```

```
iperf3 -c 127.0.0.1 -u -b 30M -t 10 -i 1
```

```
iperf3 -c 127.0.0.1 -u -b 40M -t 10 -i 1
```

```
iperf3 -c 127.0.0.1 -u -b 50M -t 10 -i 1
```

```
iperf3 -c 127.0.0.1 -u -b 60M -t 10 -i 1
```

III. Plot the Graph

```
python3 plot.py
```

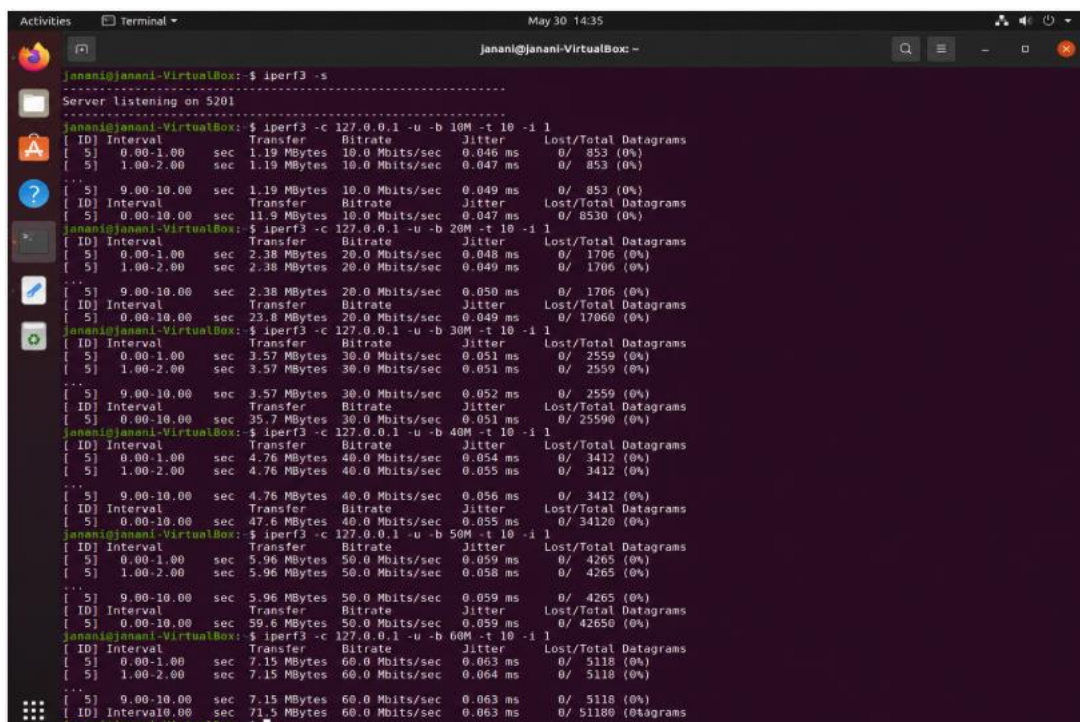
iv. Python Library Used

```
import matplotlib.pyplot as plt
```

6. Observation

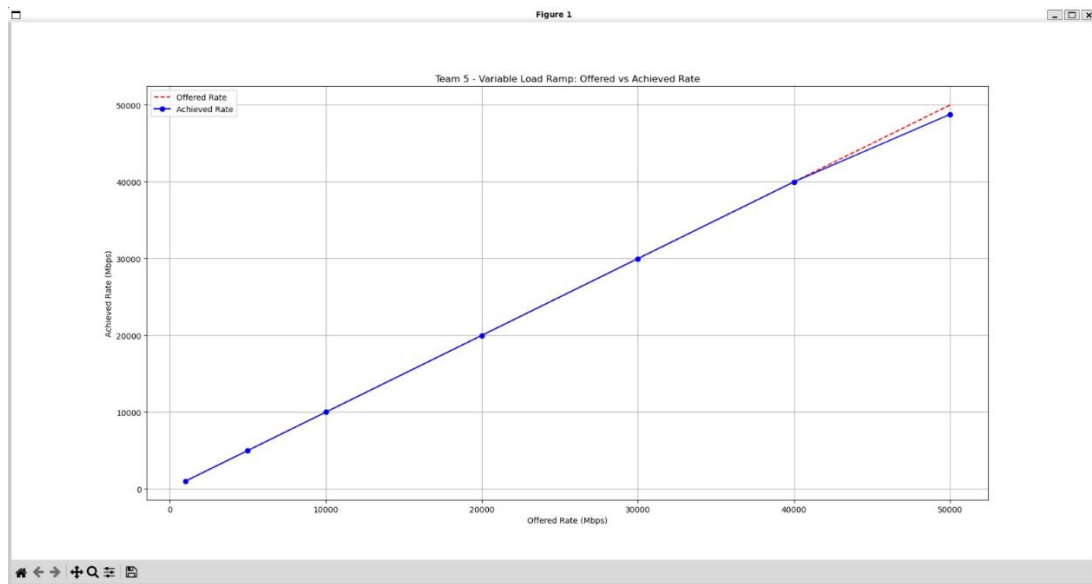
- The UDP connection between the client and server was established successfully.
- The offered traffic increased gradually from 10 Mbps to 60 Mbps in six stages.
- The achieved throughput closely followed the target bandwidth throughout the experiment.
- Small variations were observed during each transition because of normal processing delay.
- No packet loss was observed during the experiment.
- Jitter remained very low, indicating stable network performance.
- The throughput graph clearly shows a step-wise increase corresponding to the offered bandwidth.
- The network successfully handled the increasing traffic load without significant performance degradation.
- Overall, the system demonstrated reliable and consistent throughput under variable traffic conditions.

7.RESULT:



```
janani@janani-VirtualBox:~$ iperf3 -s
Server listening on 5201

janani@janani-VirtualBox:~$ iperf3 -c 127.0.0.1 -u -b 10M -t 10 -i 1
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-1.00    sec 1.19 MBytes 10.0 Mbits/sec 0.046 ms    0/ 852 (0%)
[ 5] 1.00-2.00    sec 1.19 MBytes 10.0 Mbits/sec 0.047 ms    0/ 853 (0%)
...
[ 5] 9.00-10.00   sec 1.19 MBytes 10.0 Mbits/sec 0.049 ms    0/ 853 (0%)
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-10.00   sec 11.9 MBytes 10.0 Mbits/sec 0.047 ms    0/ 8530 (0%)
janani@janani-VirtualBox:~$ iperf3 -c 127.0.0.1 -u -b 20M -t 10 -i 1
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-1.00    sec 2.38 MBytes 20.0 Mbits/sec 0.048 ms    0/ 1706 (0%)
[ 5] 1.00-2.00    sec 2.38 MBytes 20.0 Mbits/sec 0.049 ms    0/ 1706 (0%)
...
[ 5] 9.00-10.00   sec 2.38 MBytes 20.0 Mbits/sec 0.050 ms    0/ 1706 (0%)
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-10.00   sec 23.8 MBytes 20.0 Mbits/sec 0.049 ms    0/ 17060 (0%)
janani@janani-VirtualBox:~$ iperf3 -c 127.0.0.1 -u -b 30M -t 10 -i 1
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-1.00    sec 3.57 MBytes 30.0 Mbits/sec 0.051 ms    0/ 2559 (0%)
[ 5] 1.00-2.00    sec 3.57 MBytes 30.0 Mbits/sec 0.051 ms    0/ 2559 (0%)
...
[ 5] 9.00-10.00   sec 3.57 MBytes 30.0 Mbits/sec 0.052 ms    0/ 2559 (0%)
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-10.00   sec 35.7 MBytes 30.0 Mbits/sec 0.051 ms    0/ 25590 (0%)
janani@janani-VirtualBox:~$ iperf3 -c 127.0.0.1 -u -b 40M -t 10 -i 1
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-1.00    sec 4.76 MBytes 40.0 Mbits/sec 0.054 ms    0/ 3412 (0%)
[ 5] 1.00-2.00    sec 4.76 MBytes 40.0 Mbits/sec 0.055 ms    0/ 3412 (0%)
...
[ 5] 9.00-10.00   sec 4.76 MBytes 40.0 Mbits/sec 0.056 ms    0/ 3412 (0%)
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-10.00   sec 47.6 MBytes 40.0 Mbits/sec 0.055 ms    0/ 34120 (0%)
janani@janani-VirtualBox:~$ iperf3 -c 127.0.0.1 -u -b 50M -t 10 -i 1
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-1.00    sec 5.96 MBytes 50.0 Mbits/sec 0.059 ms    0/ 4265 (0%)
[ 5] 1.00-2.00    sec 5.96 MBytes 50.0 Mbits/sec 0.058 ms    0/ 4265 (0%)
...
[ 5] 9.00-10.00   sec 5.96 MBytes 50.0 Mbits/sec 0.059 ms    0/ 4265 (0%)
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-10.00   sec 59.6 MBytes 50.0 Mbits/sec 0.059 ms    0/ 42650 (0%)
janani@janani-VirtualBox:~$ iperf3 -c 127.0.0.1 -u -b 60M -t 10 -i 1
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-1.00    sec 7.15 MBytes 60.0 Mbits/sec 0.063 ms    0/ 5118 (0%)
[ 5] 1.00-2.00    sec 7.15 MBytes 60.0 Mbits/sec 0.064 ms    0/ 5118 (0%)
...
[ 5] 9.00-10.00   sec 7.15 MBytes 60.0 Mbits/sec 0.063 ms    0/ 5118 (0%)
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Tot. Datagrams
[ 5] 0.00-10.00   sec 71.5 MBytes 60.0 Mbits/sec 0.063 ms    0/ 51180 (0%)
janani@janani-VirtualBox:~$
```



8. Conclusion

The Variable Load Ramp experiment was successfully carried out using iPerf3 on Ubuntu Linux. The offered traffic was increased progressively from 10 Mbps to 60 Mbps, and the achieved throughput closely followed the target bandwidth throughout the experiment. The throughput graph confirms stable network performance with minimal fluctuations, negligible jitter, and no packet loss. The results demonstrate that the network can efficiently adapt to increasing traffic loads while maintaining reliable data transmission. This experiment provides a clear understanding of throughput behavior under dynamically changing network conditions and serves as a useful reference for evaluating network performance in future Open5GS-based experiments.

9. References

1. Open5GS Documentation <https://open5gs.org/open5gs/docs/guide/02-building-open5gs-from-sources/>
2. iPerf3 Documentation <https://iperf.fr/>
3. Open5GS GitHub Repository <https://github.com/open5gs/open5gs>

